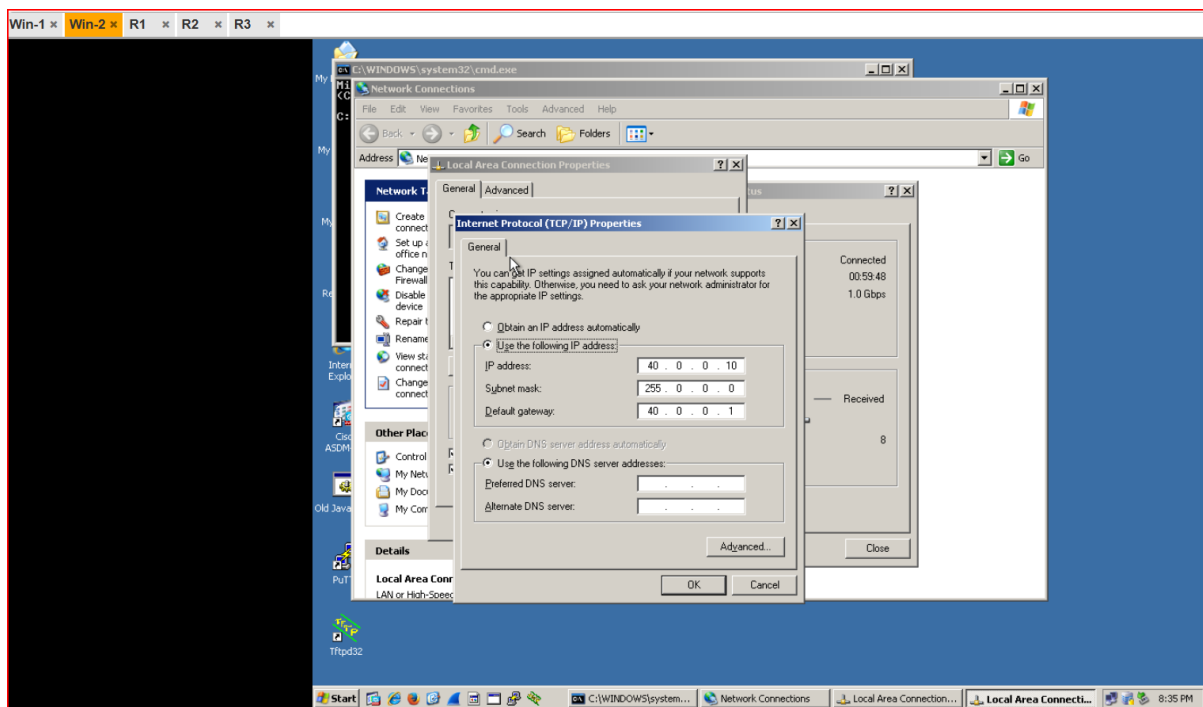
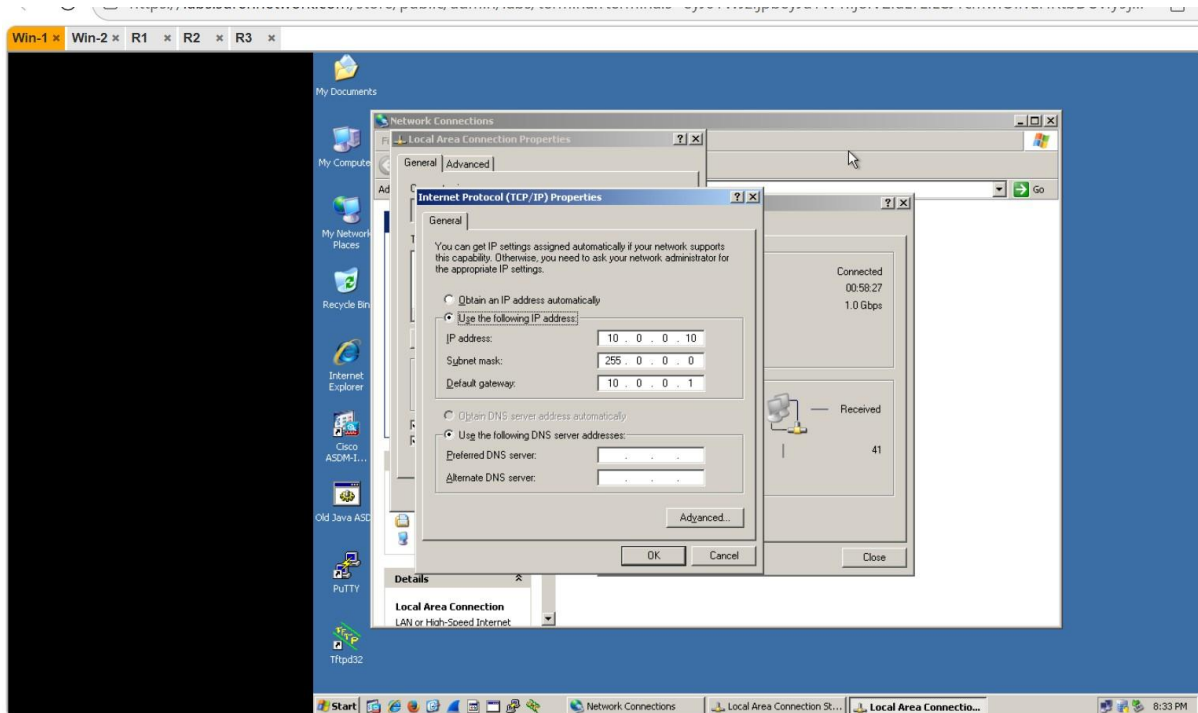
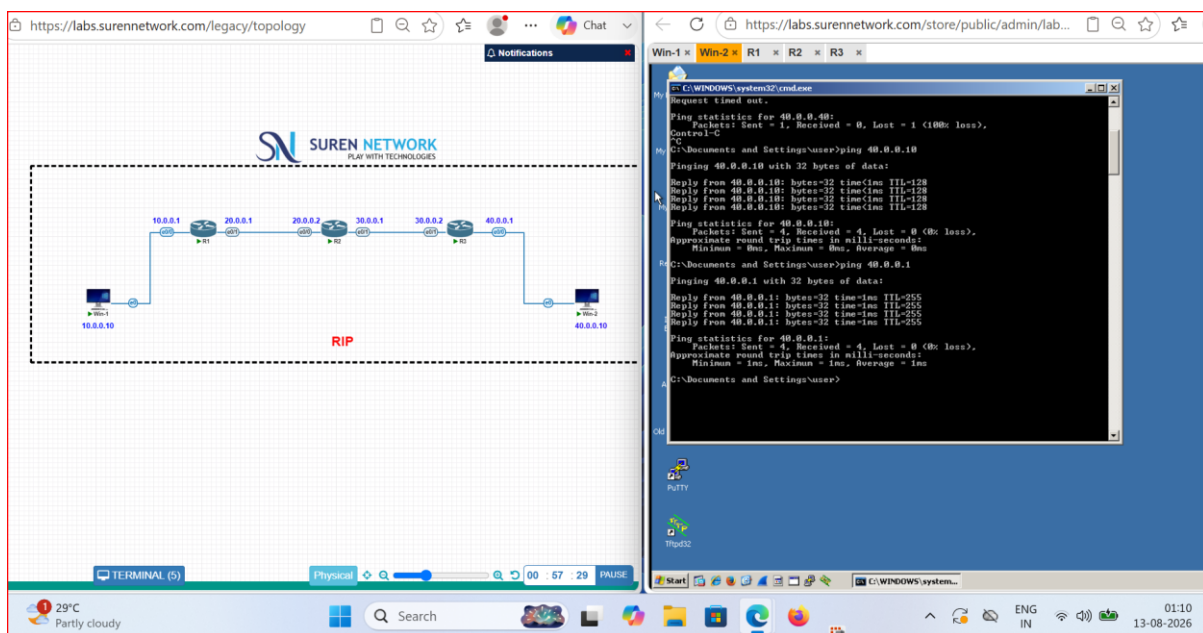
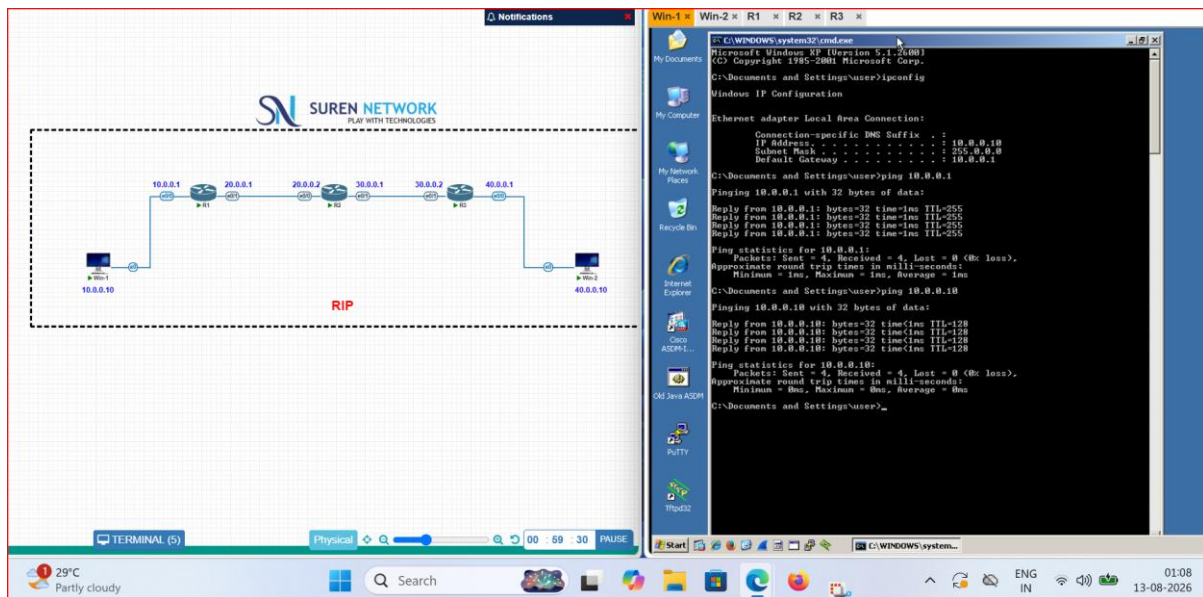


Lab Static Router Configuration

PC1 ip address, subnet 10.0.0.10 ,255.0.0.0 ,PC2-40.0.0.10 255.0.0.0 configuration completed





Router1,2,3,between three router configuration format as below

Router R1 open enter

>user mode to go # privileges mode

Enable R1#

Configure terminal

R1#Create username password

R1#Username ccan password 1234

R1#Line console 0

R1#Login local

R1#Password secret 1234

Enable talent

R1#Line vty 0 4

R1#Transport input telnet

R1#Password telnet

R1#Login local

Exit

Save write memory

Exit

Back to configure to uem mode

Check show ip interface Ethernet 0/0,Ethernet 0/1 assigned mention ip address all Router for 1,2,3

Check for status and protocols down or up status

After that checking route ip to designation ip ping or not (if not ping check the route table route & add route for nearby hub location ip address net mask network and host)

Win-1 * Win-2 * R1 * R2 * R3 *

User Access Verification

Username: ccna

Password:

R1>en

R1>enable

Password:

R1#show ip inter

R1#show ip interface br

R1#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
Ethernet0/0	10.0.0.1	YES	manual	up	up
Ethernet0/1	20.0.0.1	YES	manual	up	up
Ethernet0/2	unassigned	YES	TFTP	down	down
Ethernet0/3	unassigned	YES	TFTP	down	down

R1#show ip runin

R1#show ip rout

R1#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP

a - application route

+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks

C 10.0.0.0/8 is directly connected, Ethernet0/0

L 10.0.0.1/32 is directly connected, Ethernet0/0

20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks

```
Win-1 x Win-2 x R1 x R2 x R3 x
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

  10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    10.0.0.0/8 is directly connected, Ethernet0/0
L    10.0.0.1/32 is directly connected, Ethernet0/0
  20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    20.0.0.0/8 is directly connected, Ethernet0/1
L    20.0.0.1/32 is directly connected, Ethernet0/1
  30.0.0.0/24 is subnetted, 1 subnets
S    30.0.0.0 [1/0] via 20.0.0.2
  40.0.0.0/24 is subnetted, 1 subnets
S    40.0.0.0 [1/0] via 20.0.0.2
R1#show runin
R1#show runnin
R1#show running-config
Building configuration...

Current configuration : 1414 bytes
!
! Last configuration change at 00:53:30 IST Thu Aug 13 2026 by ccna
!
version 15.7
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname R1
!
boot-start-marker
boot-end-marker
```

```
R2#
R2#
R2#conf
R2#configure ter
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#
R2(config)#ip route 10.0.0.0 255.255.255.0 20.0.0.1
R2(config)#ip route 40.0.0.0 255.255.255.0 30.0.0.2
R2(config)#exit
R2#
*Aug 12 19:27:48.928: %SYS-5-CONFIG_I: Configured from console by ccna on console
R2#write me
R2#write memory
Building configuration...
[OK]
R2#show ip r
R2#show ip rout
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        a - application route
        + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

    10.0.0.0/24 is subnetted, 1 subnets
S       10.0.0.0 [1/0] via 20.0.0.1
    20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.0/8 is directly connected, Ethernet0/0
L       20.0.0.2/32 is directly connected, Ethernet0/0
```

Win-1 x Win-2 x R1 x R2 x R3 x

```
R3>enable
Password:
R3#show ip inter
R3#show ip interface br
R3#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
Ethernet0/0              40.0.0.1        YES manual up          up
Ethernet0/1              30.0.0.2        YES manual up          up
Ethernet0/2              unassigned      YES TFTP  down        down
Ethernet0/3              unassigned      YES TFTP  down        down
R3#show ip r
R3#show ip rout
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

    10.0.0.0/24 is subnetted, 1 subnets
S       10.0.0.0 [1/0] via 30.0.0.1
    20.0.0.0/24 is subnetted, 1 subnets
S       20.0.0.0 [1/0] via 30.0.0.1
    30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       30.0.0.0/8 is directly connected, Ethernet0/1
L       30.0.0.2/32 is directly connected, Ethernet0/1
    40.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       40.0.0.0/8 is directly connected, Ethernet0/0
L       40.0.0.1/32 is directly connected, Ethernet0/0
R3#
```



```
Win-1 x Win-2 x R1 x R2 x R3 x
C:\WINDOWS\system32\cmd.exe
C:\Documents and Settings\user>ping 10.0.0.10
Pinging 10.0.0.10 with 32 bytes of data:
Reply from 10.0.0.10: bytes=32 time<1ms TTL=128
Reply from 10.0.0.10: bytes=32 time<1ms TTL=128
Reply from 10.0.0.10: bytes=32 time<1ms TTL=128
Reply from 10.0.0.10: bytes=32 time<1ms TTL=128
Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\Documents and Settings\user>route print
=====
Interface List
0x1 ..... MS TCP Loopback interface
0x2 ...50 cf cb 01 aa 00 ..... Intel(R) PRO/1000 MT Network Connection - Packet
Scheduler Miniport
=====
Active Routes:
Network Destination    Netmask          Gateway          Interface        Metric
0.0.0.0                0.0.0.0          10.0.0.1         10.0.0.10        10
10.0.0.0               255.0.0.0        10.0.0.10       10.0.0.10        10
10.0.0.10             255.255.255.255  127.0.0.1       127.0.0.1        10
10.255.255.255        255.255.255.255  10.0.0.10       10.0.0.10        10
127.0.0.0             255.0.0.0        127.0.0.1       127.0.0.1        1
224.0.0.0            240.0.0.0        10.0.0.10       10.0.0.10        10
255.255.255.255      255.255.255.255  10.0.0.10       10.0.0.10        1
Default Gateway:      10.0.0.1
=====
Persistent Routes:
None
C:\Documents and Settings\user>
```

Finally, route directly connected same network ping different network assigned for the router route

Assigned for

Router 1 to router 2

Router 2 to router 3

Router 3 to pc 2 communication reached the source ip to designation ip ping

